THE INSTITUTION OF ENGINEERS (INDIA)
KUWAIT CHAPTER • TECHNICAL ARCHITECTURE REPORT

TECHNICAL BRIEF
Google Auth Resolution

Google Authentication Error: Root Cause & Solution Guide

Target System: IEI Mobile Application (React Native / Expo SDK 57 & NestJS Backend) • Date: October 2026

1. EXECUTIVE SUMMARY

When tapping **"Continue with Google"** in the IEI Kuwait Mobile Application running on an Android device via Expo Go, a browser window opened to `accounts.google.com` displaying:

> **Google Error Encountered:**
> *"404. That's an error. The requested URL was not found on this server. That's all we know."*

This technical document explains the exact root cause behind this behavior and outlines two distinct solutions: an **immediate zero-setup testing path** for application demonstrations, and the **step-by-step production configuration** for Google Cloud Console.

2. ROOT CAUSE ANALYSIS (DEEP-DIVE)

Google's OAuth 2.0 authorization server (`accounts.google.com/o/oauth2/v2/auth`) validates incoming authorization requests against strict security guidelines. Two fatal conditions occurred simultaneously:

OAuth Parameter	Value Sent by Mobile App	Google Server Response
client_id	`1082348721389-dev.apps.googleusercontent.com` (Placeholder)	`invalid_client` The OAuth client was not found in Google Cloud.
redirect_uri	`exp://192.168.x.x:8081` (Expo Go scheme)	`invalid_request` Non-compliant custom scheme; violates Google OAuth 2.0 security policy.

> **Why Did Google Show a "404 Not Found" Instead of a Helpful Screen?**
> When Google detects security and protocol violations from an external mobile agent, it issues an HTTP 302 redirecting to its internal error URL (`https://accounts.google.com/signin/oauth/error?authError=...`). Because this private URL is invoked inside an isolated mobile Chrome Custom Tab with no active session or authorization context, Google fails to resolve the session state and returns its standard generic fallback: **404. That's an error.**

3. SOLUTION A: IMMEDIATE DEMO TESTING (NO SETUP REQUIRED)

If you need to test the application, navigate screens, or perform live client demonstrations immediately without waiting for Google Cloud Console provisioning:

Direct Database Authentication Path:

Tap **"Login with Email / Mobile"** located right below the OR divider on the login screen. You can log in using any of the pre-reconciled live demo members in your PostgreSQL database.

Member ID	Mobile Number	Designation / Post	Permissions & Access
IEI9002	`00000002`	General Secretary	ADMIN Event check-in scanning, member reviews
IEI9003	`00000003`	Chapter Chairman	ADMIN Complete executive view, governance
IEI9004	`00000004`	EC Volunteer	VOLUNTEER Event Door Check-in & scanning
IEI9201	`00009201`	Active Member Alpha	MEMBER Digital ID card, profile, declarations
IEI9202	`00009202`	Active Member Beta	MEMBER Digital ID card, profile, declarations

4. SOLUTION B: PRODUCTION GOOGLE CLOUD CONFIGURATION

To enable native Google Sign-In with real Google accounts, follow this 4-step configuration in the Google Cloud Console:

1. **Create / Select Project in Google Cloud Console:**
   Visit console.cloud.google.com, create a project named `IEI-Kuwait-App`, and navigate to **APIs & Services > OAuth consent screen**. Select *External*, enter App Name (`IEI Kuwait`), Support Email, and Developer Contact.
2. **Create Android OAuth 2.0 Client ID:**
   Go to **APIs & Services > Credentials > Create Credentials > OAuth client ID**:
   - Application type: **Android**
   - Package name: `org.iei.kuwait` (matches `app.json`)
   - SHA-1 certificate fingerprint: Obtain by running `npx expo credentials:manager` or extracting from your Android debug/release keystore.
3. **Create Web Application Client ID (For Expo Go & Server Verification):**
   Create a second credential with Application type: **Web application**:
   - Name: `IEI Mobile Web Client`
   - Authorized JavaScript origins: `https://auth.expo.io`
   - Authorized redirect URIs: `https://auth.expo.io/@anonymous/IEI-Mobile` and `ieimobile://oauthredirect`
4. **Configure Environment Variables in Mobile App:**
   Paste the generated Client IDs into **`IEI-Mobile/.env`**:

```
# IEI-Mobile/.env
EXPO_PUBLIC_GOOGLE_WEB_CLIENT_ID="[YOUR_CLIENT_ID].apps.googleusercontent.com"
EXPO_PUBLIC_GOOGLE_ANDROID_CLIENT_ID="[YOUR_ANDROID_CLIENT_ID].apps.googleusercontent.com"
EXPO_PUBLIC_GOOGLE_IOS_CLIENT_ID="[YOUR_IOS_CLIENT_ID].apps.googleusercontent.com"
```

5. BACKEND CRYPTOGRAPHIC VERIFICATION ARCHITECTURE

Once the user selects their account in the official Google prompt, the client sends the cryptographic `idToken` to the IEI Backend endpoint `POST /auth/google`. The backend verifies the identity token securely:

```
Mobile App (Continue with Google)
      ↓ (Client ID + redirectUri)
Google OAuth Server (User Authenticates)
      ↓ (Cryptographic Google ID Token returned)
IEI Backend: POST /auth/google
      ↓
```

```
Google Tokeninfo API (Verify Signature, aud, exp, sub)
      ↓
Match or Create AuthIdentity (provider: GOOGLE, providerUserId: sub)
      ↓
Issue 7-Day Signed IEI JWT Session
      ↓
Mobile App Stores Session & Redirects to Member Dashboard
```

6. CODE SAFEGUARDS IMPLEMENTED

To prevent any user from ever seeing Google's raw 404 screen again, the following safeguards are in place:

- **Client ID Guard:** If no valid Google Client ID is configured in the environment, the app stops the browser from opening and displays a clear message directly on the login screen.
- **Compliant Redirect URI:** `makeRedirectUri({ scheme: 'ieimobile', path: 'oauthredirect' })` is registered to prevent transmission of non-compliant `exp://` URIs.
- **Zero Breaking Changes:** All existing design tokens, fonts, logos, guest browsing, and membership application links remain intact.
